Devita Syifa 1A TTE

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No.

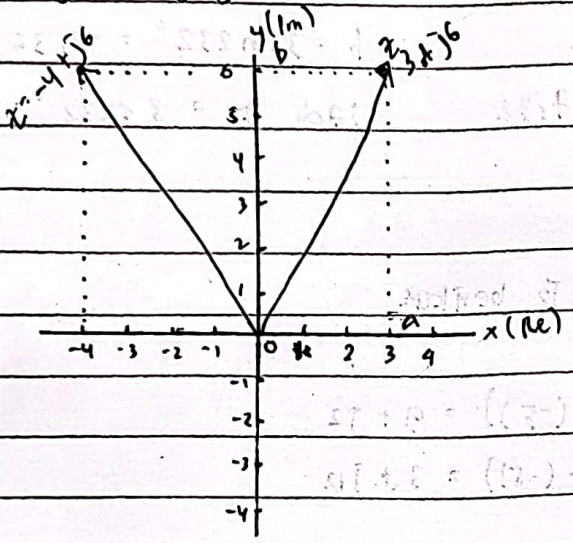
Date: Bil. Imajiner

1. Sederhanakan bilangan imajiner berikut!

a. $j^3 \cdot 1 \cdot j = -j$

b. $j^4 = -j : j = \sqrt{-1}$

2.



$a = -4 + j6$

$b = 3 + j6$

3. Tulislah bentuk sekawan (conyugate) dan lawan bilangan kompleks berikut!

a. $4 + j \Rightarrow \bar{z} = \overline{4 + j} = 4 - j$, lawan $= -4 - j$

b. $3 - j2 \Rightarrow \bar{z} = \overline{3 - j2} = 3 + j2$, lawan $= -3 + j2$

4. Nyatakan bilangan kompleks berikut ke dalam bentuk polar!

a. $-1 + j \Rightarrow z = -1 + j \rightarrow Re(z) = -1$ dan $Im(z) = 1 \Rightarrow$ kuadran II

dengan $r = \sqrt{(-1)^2 + (1)^2} = \sqrt{2}$ dan $\theta = \arctan \frac{1}{-1} = -45°$

$180° - 45° = 135°$ pada kuadran 2 $\Rightarrow \sqrt{2} < 135°$

d. $-1 + j \Rightarrow z = -1 + j \rightarrow Re(z) = -1$ dan $Im(z) = 1 \Rightarrow$ kuadran II

dengan $r = \sqrt{(-1)^2 + (1)^2} = \sqrt{2}$ dan $\theta = \arctan \frac{1}{-1} = -45°$

$180° - 45° = 135°$ pada kuadran 2 $\Rightarrow \sqrt{2} < 135°$

5. Nyatakan bilangan kompleks berikut dalam bentuk eksponen!

b. $1 + j\sqrt{3}$ terletak di kuadran I, dengan $r = \sqrt{1^2 + (\sqrt{3})^2} = \sqrt{4} = 2$.

$\theta = \tan^{-1} \frac{\sqrt{3}}{1} = 60° = \frac{\pi}{3}$, jadi $z = 2e^{j\frac{\pi}{3}}$

c. $\sqrt{2} - j\sqrt{2}$ terletak di kuadran IV dengan $r = \sqrt{\sqrt{2}^2 + (-\sqrt{2})^2} = 2$

$\theta = \tan^{-1} \frac{-\sqrt{2}}{\sqrt{2}} = -45°$, karena di kuadran IV, maka $\theta = 360° - 45° = 315° = \frac{7\pi}{4}$

maka $\rightarrow 2e^{j\frac{7\pi}{4}}$

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6. Nyatakan dalam bentuk cartesian

d. $7,32 \angle -270^\circ$

$a = 7,32 \cos(-270^\circ) = 0$

$b = 7,32 \sin(-270^\circ) = 7,32$

Jadi $z = 7,32 \angle -270^\circ = j7,32$

e. $3 \angle 232^\circ$

$a = 3 \cos 232^\circ = -1,84$

$b = 3 \sin 232^\circ = -2,36$

Jadi $z = 3 \angle 232^\circ = -1,84 - j2,36$

Bagian B

1. Sederhanakan dalam bentuk $a + jb$ berikut

d. $(6 + j7) + (3 - j5)$

$z_1 + z_2 = (6+3) + j(7 + (-5)) = 9 + j2$

$z_1 - z_2 = (6-3) + j(7 - (-5)) = 3 + j12$

b. $(2 - \sqrt{-4}) + (-3 + \sqrt{-16})$

$= (2 - j2) + (-3 + j4)$

$z_1 + z_2 = (2 + (-3)) + j(-2+4) = -1 + j2$

$z_1 - z_2 = (2 - (-3)) + j(-2-4) = 5 - j6$

2. Operasikan perkalian berikut dalam cartesian

b. $(2-j)(3+j)$

$z_1 z_2 = (2.3 - (-1).(1)) + j(2(1) + (-1)(3)) = 7 - j$

d. $(3+j2)(3-j2)$

$z_1 z_2 = (3.3 - (2)(-2)) + j(3(-2) + (2)3) = 13$

3. Operasikan perkalian berikut dalam bentuk cartesian!

a. $\frac{3+j4}{1-j2} = \frac{(3-8) + j(4-(-6))}{1^2 + (-2)^2}$

$= \frac{-5 + j10}{5} = -1 + j2$

d. $\frac{1}{(2+j)(1+j2)}$

$= \frac{1}{(2.1 - 1(2)) + j(2.2 + 1.1)}$

$= \frac{1}{j5}$

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4. Selesaikan :

a. $(12+j)-(4+j5)$

$z_1 z_2 = (12-4) + j(1-5) = 8 - j4$

b. $(3+j2)(4-j3)$

$z_1 z_2 = (3 \cdot 4 - 2(-3)) + j(3(-3) + 2(4)) = 18 - j$

5. dik : $z_1 = 3 + j2$, $z_2 = 2 - j3$, $z_3 = -4 + j$, $z_4 = -3 - j2$

Hitung :

a. $z_1 z_3 = (3 + j2)(-4 + j)$

$= (3(-4)) - 2(1)) + j(3 \cdot 1 + 2(-4))$

$= -14 - j5$

d. $\frac{z_1 - z_4}{z_2 + z_3} = \frac{(3+j2) - (-3-j2)}{(2-j3) + (-4+j)}$

$= \frac{(3-(-3)) + 3(2-(-2))}{(2+(-4)) + j(-3+1)}$

$= \frac{6 + j4}{-2 - j2}$

$= \frac{(-12 - 8) + j(-8 - (-12))}{(-2)^2 + (-2)^2}$

$= \frac{-20 + j4}{8} = \frac{-5 + j}{2}$

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Latihan 2

1. b. $(2-j)(3+j)$

$z_1 z_2 = 6 - j3 + j2 - j^2 = 6 - j - (-1) = 7 - j$

c. $(-2+j3)(-1-j2)$

$z_1 z_2 = 2 - j3 + j4 - 6j^2 = 2 + j - 6(-1) = 8 + j$

2. a. $\frac{3+j4}{1-j2} = \frac{3+j4}{1-j2} \cdot \frac{1+j2}{1+j2} = \frac{3+j4+j6+8j^2}{1+j^2-j2-4j^2} = \frac{-5+j10}{5}$

3. b. $z_1 z_3 = (3+j2)(-4+j) = -12-8j+3j+2j^2 = -(14-5j)$

d. $\frac{z_2}{z_4} = \frac{2-j3}{-3-j2} \cdot \frac{-3+j2}{-3+j2} = \frac{-6+9j+4j-6j^2}{9+6j-6j-4j^2} = \frac{13j}{13} = j$

4. a. $(a-j2) - (7+jb) = -4-j8$

$(a-7) + j(-2-b) = -4-j8$

$a-7 = -4 \qquad -2-b = -8$

$a = 3 \qquad b = 6$

b. $(a+jb) - (\sqrt{4} - \sqrt{-16} = 1-j6$

$(a+jb) - (2-j4) = 1-j6$

$(a-2) + j(b+4) = 1-j6$

$a-2 = 1 \qquad b+4 = -6$

$a = -1 \qquad b = -10$

5. a. $3\angle 23^\circ \times 4\angle 46^\circ$

$z_1 z_2 = (3 \times 4) \angle (23+46)^\circ = 12\angle 69^\circ$

d. $\frac{7\angle 20}{3\angle 50} = (7/3)\angle(20-50)^\circ = 2.33\angle -30^\circ$

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6. a). $(1-j\sqrt{3})^2$

$z = 1-j\sqrt{3} \rightarrow z = 2 \angle \frac{\pi}{3}$

$z^2 = (1-j\sqrt{3})^2 \Rightarrow (2 \angle \frac{\pi}{3})^2 = 4 \angle \frac{2}{3}\pi$

b). $(-1+j)^3$

$z = -1+i \Rightarrow z = 1 \angle \pi/4$

$z^3 = (-1+i)^3 = (1 \angle \frac{\pi}{4})^3 = 13 \angle \frac{3\pi}{4} = 1 \angle \frac{3}{4}\pi$

7. d). $8 \angle 110°\ (z)$

$z_0 = \frac{8^{1/2} \angle ((110° + 2(0))\pi)}{2} = 2\sqrt{2} \angle 55°$

$z_1 = \frac{8^{1/2} \angle (110° + 2(1)\pi)}{2} = 2\sqrt{2} \angle 235°$

a). $2+j \rightarrow \sqrt{5} \angle \pi/4\ (z)$

$z_0 = (5^{\frac{1}{2}})^{\frac{1}{2}} \angle \frac{\pi/4 + 2(0)\pi)}{2} = 5^{1/4} \angle \frac{9\pi}{8}$

8. a). $z_1 z_2$

(i) $z_1 = 3 \angle \pi/3,\ z_2 = 5 \angle \pi/4$

$z_1 . z_2 = (3 . 5) \angle (\pi/3 + \pi/4)$

$= 15 \angle 7\pi/12$

(ii) $z_1 = \sqrt{6}+\sqrt{3},\ z_2 = 2\sqrt{5}+j$

$z_1 . z_2 = (\sqrt{6}+\sqrt{3})(2\sqrt{6}+j)$

$= 12 + j\sqrt{6} + j6\sqrt{6} + j^2\sqrt{3}$

$= 12 - \sqrt{3} + j7\sqrt{6}$

c). z_1/z_2

(i). $z_1 = 3 \angle \pi/3,\ z_2 = 5 \angle \pi/4$

$z_1/z_2 = \frac{3}{5} \angle (\frac{\pi}{3} \frac{\pi}{4})$

$= \frac{3}{5} \angle \frac{\pi}{12}$

(ii) $z_1 = \sqrt{6}+j\sqrt{3},\ z_2 = 2\sqrt{5}+j$

$z = \frac{\sqrt{6}+j\sqrt{3}}{2\sqrt{6}+j} = \frac{\sqrt{6}+j\sqrt{3}}{2\sqrt{6}+j} \cdot \frac{2\sqrt{6}-j}{2\sqrt{6}-j}$

$= \frac{12 - \sqrt{3} + j7\sqrt{6}}{25}$

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